

STM32CubeIDE installation guide

... and tutorial on how to import
the **FW_template**

Introduction

This installation guide for **STM32CubeIDE** gives directions on how to install software on each of the operating systems it supports. It is primarily intended to software developers or system administrators who are about to install the **STM32CubeIDE** product.

This installation guide covers the following topics:

- System requirements
- Important information
- STM32CubeIDE installation (Windows)
- STM32CubeIDE installation (Linux)
- STM32CubeIDE installation (macOS)
- Update an STM32CubeIDE installation
- Uninstall STM32CubeIDE (Windows)
- Uninstall STM32CubeIDE (Linux)
- Uninstall STM32CubeIDE (macOS)

ET.PM3 and ET.PM4

Please follow the procedure
in the following textboxes.

Download IDE from <https://www.st.com/en/development-tools/stm32cubeide.html>

Follow the instructions on the following pages to install the IDE.

After that, the textbox below shows how to import, compile and run the demo_code.

Start "STM32CubeIDE"

Change the workspace folder if desired. Remember the path for later.

Close the "Welcome to STM32CubeIDE" page.

Close the program "STM32CubeIDE"

Download the archive "FW_template_xxxxxxx.zip" to a local folder on your pc.

Connect the evaluation board "STM32F429I-DISC1" via "USB ST-LINK" on top of the board.

Start "STM32CubeIDE" again

Menu: File -> Import ... -> General -> Existing Project into Workspace -> Next

Select radio button: *Select Archive File*. click "Browse", Browse to downloaded zip archive, select it and click "Open" and then click "Finish"

After the import is finished, the "FW_template" appears in the "Project Explorer" on the left.

Click on "FW_template" then "Core". Here are the source files.

Menu: Project -> Build All

Menu: Run -> Debug

In the firewall dialog -> allow

In the dialog "Confirm Perspective Switch" -> "Switch"

Wait for "Download verified successfully" to appear in the console window on the bottom.

Menu: Run -> Resume

Now the demo runs on the evaluation board.

1 System requirements

STM32CubeIDE is tested and verified on the Microsoft® Windows®, Linux®, and macOS® versions listed in this chapter.

Important: Only 64-bit OS versions are supported.

STM32CubeIDE supports STM32 32-bit products based on the Arm® Cortex® processor.

Note: Arm is a registered trademark of Arm Limited (or its subsidiaries) in the US and/or elsewhere.

arm

1.1 Microsoft® Windows®

- Microsoft® Windows 10®

1.2 Linux®

- Ubuntu® 18.04
- Ubuntu® 20.04
- Fedora® 29
- Fedora® 31

Note: Linux® is a registered trademark of Linus Torvalds.
Ubuntu® is a registered trademark of Canonical Ltd.
Fedora® is a trademark of Red Hat, Inc.

1.3 macOS®

- macOS® 10.15 (Catalina)
- macOS® 11 (Big Sur)

Note: Refer to [Section 5.1](#) for possible macOS® Gatekeeper blocking issues.
macOS® is a trademark of Apple Inc., registered in the U.S. and other countries and regions.
All other trademarks are the property of their respective owners.

1.4 Hardware requirements

The following hardware requirements apply:

- 2 Gbytes of RAM minimum. 4 Gbytes of RAM recommended
- 6 Gbytes of free hard-disk space for non STM32 MPU OpenSTLinux Distribution developers, 15 Gbytes for STM32 MPU OpenSTLinux Distribution usage

2 Important information

This chapter contains important information regarding the installation of [STM32CubeIDE](#).

2.1 Product installer

The latest version of the [STM32CubeIDE](#) installer can be downloaded from the STMicroelectronics web site at www.st.com.

2.2 Installing from USB memory

It is not recommended to launch the [STM32CubeIDE](#) installer directly from a USB memory. Instead, copy the executable installation file from the USB memory to the local hard-disk drive of the computer and execute the installation from the hard-disk drive.

If an installation from a USB memory is still preferred, make sure that the USB memory is not write-protected and that there is at least 6 GBytes of free memory beyond the space occupied by the installation executable. The extra space is required for temporary files during the installation.

Caution: Do not remove the USB memory from the computer until the installation process is completely finished or the installation would fail.

2.3 Product upgrades

It is possible to install new versions of [STM32CubeIDE](#) in parallel with older versions.

3 STM32CubeIDE installation (Windows®)

This section describes how to install the STM32CubeIDE product on Microsoft® Windows®.

Note: The installation is done through a product installer. Make sure that the user account, from which the installer is launched, has administrative privileges.

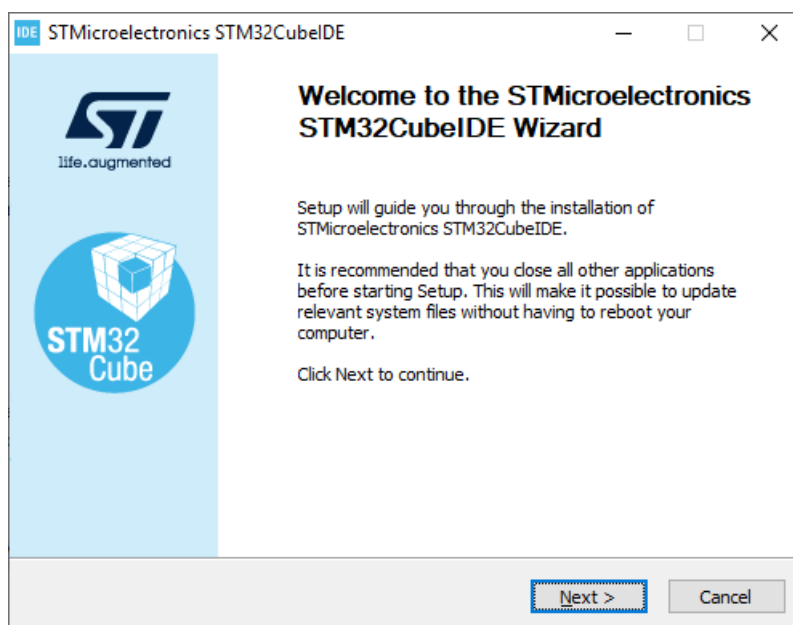
Proceed as follows:

1. Launch the product installer (.exe file)
`st-stm32cubeide_VERSION_ARCHITECTURE.exe`
where:
 - VERSION is the actual product version and build date
Example: 1.0.0_2026_20190221_1309
 - ARCHITECTURE is the architecture of the target host computer to run STM32CubeIDE
Example: x86_64

If, when launching the product installer, it reports an attempt to install a version that is already installed:

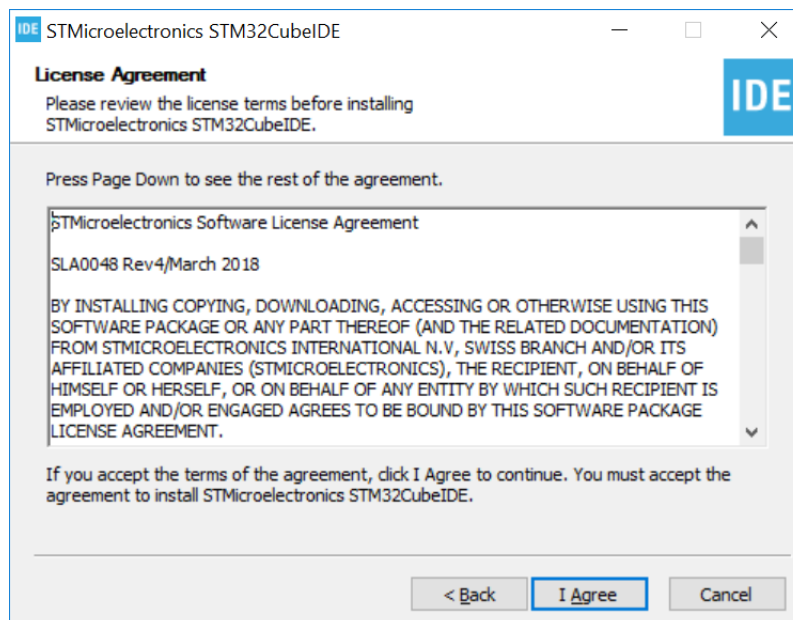
- a. Launch the register dialog by typing `regedit.exe` in the search bar
 - b. Remove registry key
`HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\STMicroelectronics\STM32CubeIDE`
2. During the installation process, the operating system may display a dialog stating: "Do you want to allow this app to make changes to your device?" with info "Verified publisher: STMicroelectronics Software AB". Accept ([YES]) to let the installer continue.
 3. Wait for the installer **Welcome** dialog to be displayed and click on [Next >].

Figure 1. Installer Welcome page (Windows®)



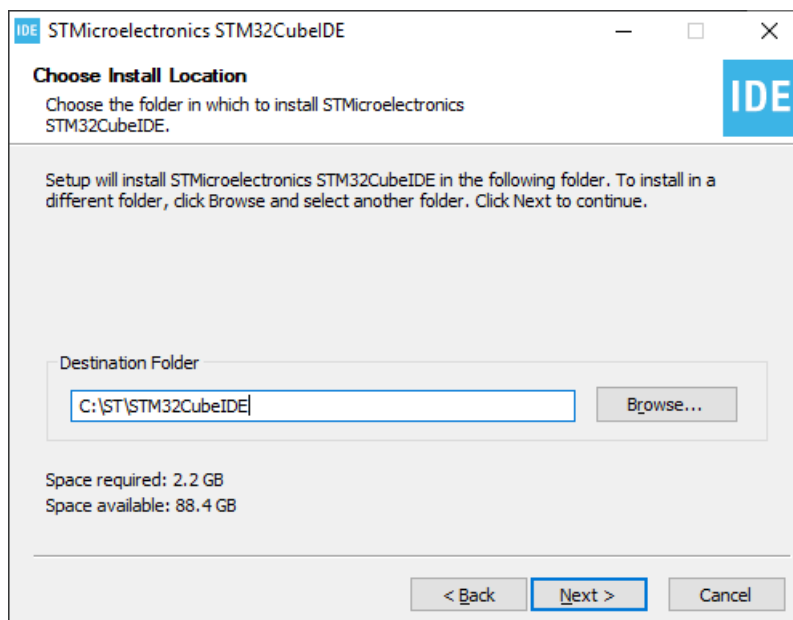
4. Read the license agreement. Click on **[I Agree]** to accept the terms of the agreement, or **[Cancel]** to abort the installation. If the agreement is accepted, the installation wizard continues.

Figure 2. License agreement dialog (Windows®)



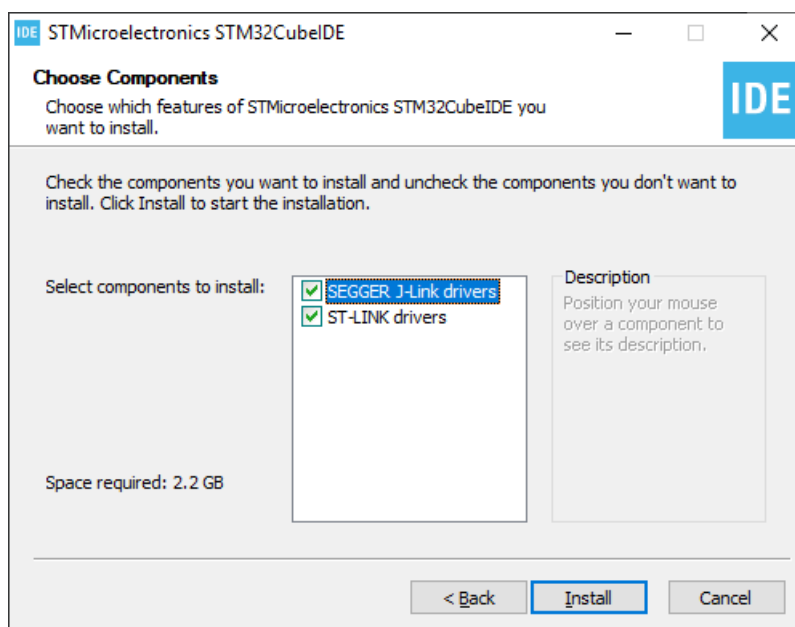
5. In this dialog, the user selects the location for the installation. It is recommended to choose a short path to avoid facing Windows® limitations with too long paths for the workspace.

Figure 3. Installer location dialog (Windows®)



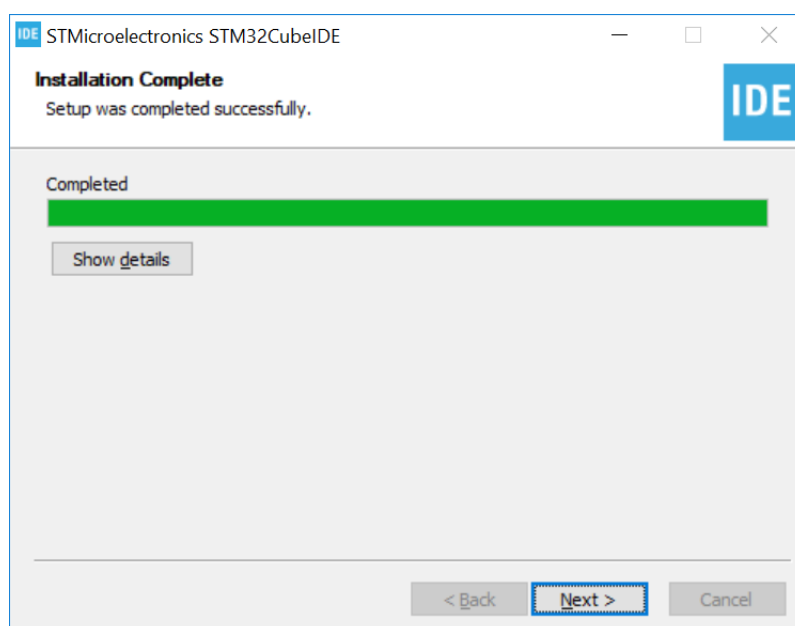
6. Wait for the **Choose Components** dialog to be displayed. Select the GDB Server components to be installed together with STM32CubeIDE. A server is needed for each type of JTAG probe used for debugging with STM32CubeIDE.

Figure 4. Selection of components dialog (Windows®)



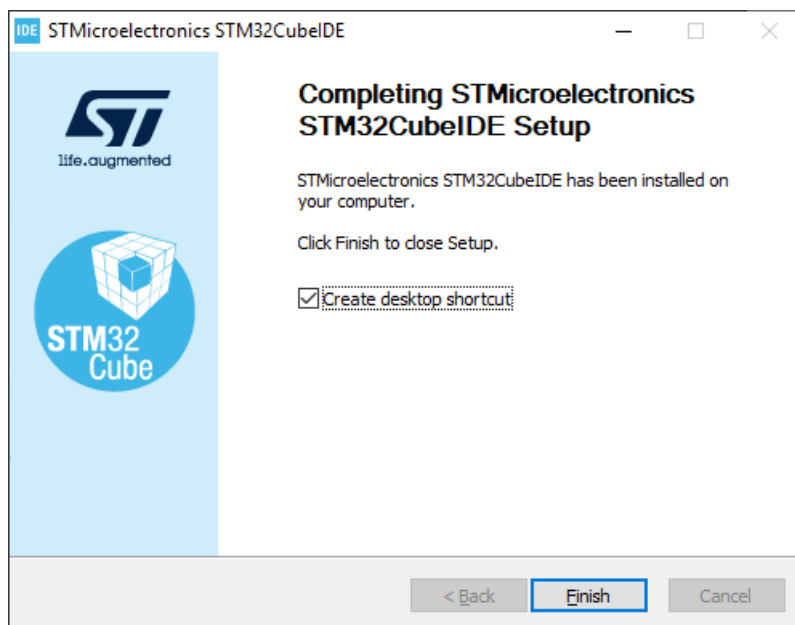
7. Click on **[Install]** to start the installation. The drivers that were selected are installed in parallel with this installation of STM32CubeIDE from here on.

Figure 5. Installation successful (Windows®)



8. Click on **[Next]** to continue to the final step of the installation process. That is a **Confirmation** dialog informing the user that the installation is finished. Once the user clicks on **[Finish]**, the installation process is complete.

Figure 6. Installation finished (Windows®)



4 STM32CubeIDE installation (Linux®)

This section describes how to install the STM32CubeIDE product on Linux®.

Note: *The installation is done through a product installer. Linux® root privileges are required to complete the installation.*

The installer comes in different bundles to suit the various Linux® distributions. The bundles are named according to:

`st-stm32cubeide_VERSION_ARCHITECTURE.PACKAGE`

where:

- **VERSION** is the actual product version and build date
Example: 1.0.0_2026_20190221_1309
- **ARCHITECTURE** is the architecture of the target host computer to run STM32CubeIDE
Example: amd64
- **PACKAGE** is the Linux® package type to be installed. The supported packages are:
 - `rpm_bundle.sh` for Fedora®/CentOS
 - `deb_bundle.sh` for Ubuntu®
 - `.sh` for generic Linux®

Proceed as follows:

1. Navigate to the location of the installer file with a command console on the host computer.
2. Enter the following command in the console window:
`sudo sh ./st-stm32cubeide_VERSION_ARCHITECTURE.PACKAGE`
where **VERSION**, **ARCHITECTURE** and **PACKAGE** must be entered after the selected Linux® package.
3. Follow the further instructions provided through the console window.

Manual installation (.rpm/.deb)

For RPM-based distributions (Red Hat®, CentOS™, SUSE®, Fedora®):

```
sudo rpm -Uhv segger-jlink-udev-rules-xxxx-linux-noarch.rpm st-stlink-udev-rules-xxxx-linux-noarch.rpm st-stlink-server-xxxx-linux-amd64.rpm st-stm32cubeide-xxxx-amd64.rpm
```

For Debian-based distributions (Debian®, Ubuntu®):

```
sudo apt-get install ./segger-jlink-udev-rules-xxxx-linux-all.deb ./st-stlink-udev-rules-xxxx-linux-all.deb ./st-stlink-server-xxxx-linux-amd64.deb ./st-stm32cubeide-xxxx-amd64.deb
```

Note: *CentOS is a trademark of Red Hat, Inc.
SUSE is a trademark of SUSE LLC or its subsidiaries or affiliates.*

5 STM32CubeIDE installation (macOS®)

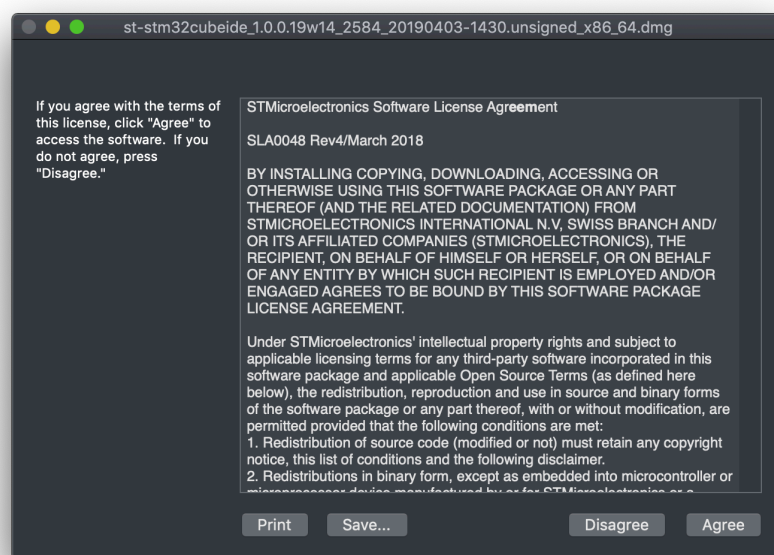
This section describes how to install the [STM32CubeIDE](#) product on macOS®.

Note: *The installation is done through a product installer. Make sure that the user account, from which the installer is launched, has administrative privileges.*

Proceed as follows:

1. Launch the product installer (.dmg file)
`st-stm32cubeide_VERSION_ARCHITECTURE.dmg`
where:
 - `VERSION` is the actual product version and build date
Example: `1.0.0_2026_20190221_13091309`
 - `ARCHITECTURE` is the architecture of the target host computer to run [STM32CubeIDE](#)
Example: `x86_64`
2. Read the license agreement. Click on **[Agree]** to accept the terms of the agreement, or **[Disagree]** to abort the installation. If the agreement is accepted, the installation wizard continues.

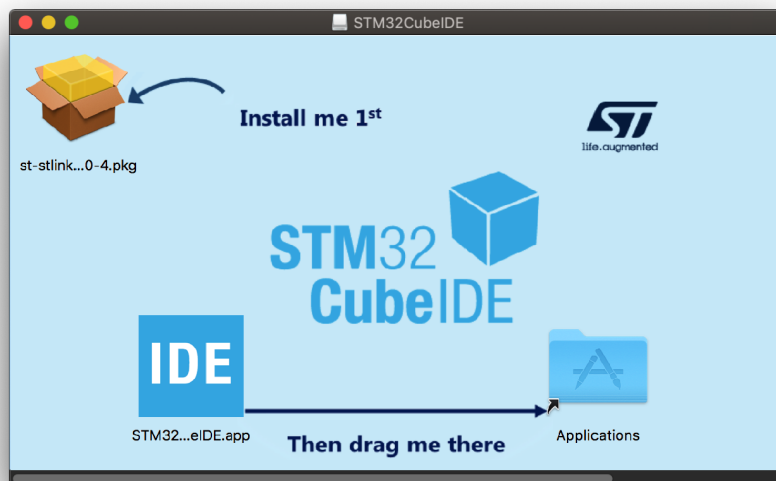
Figure 7. License agreement dialog (macOS®)



3. Wait for the installation **welcome page** to appear.

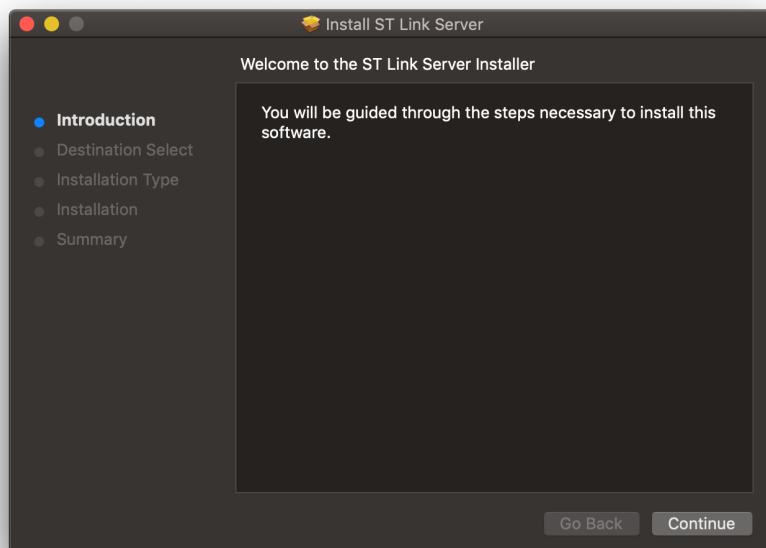
4. Double click on the .pkg file indicated with the curved arrow and text stating “Install me 1st”.

Figure 8. Installation welcome page (macOS®)



5. This installation is required and installs the *ST Link Server*. Click on the [Continue] button.

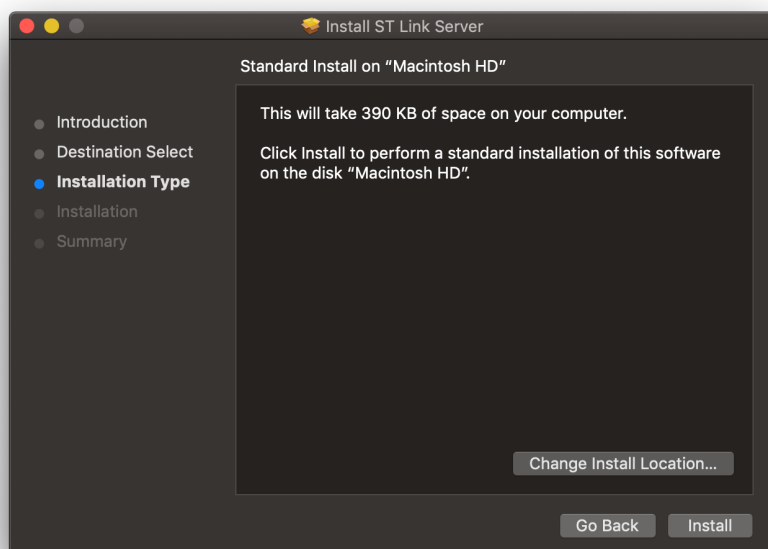
Figure 9. ST Link Server welcome page (macOS®)



6. In this dialog, select the location for the installation of the *ST Link Server*.
7. Click on [Install]. Log in to authorize the installation if prompted to do so.

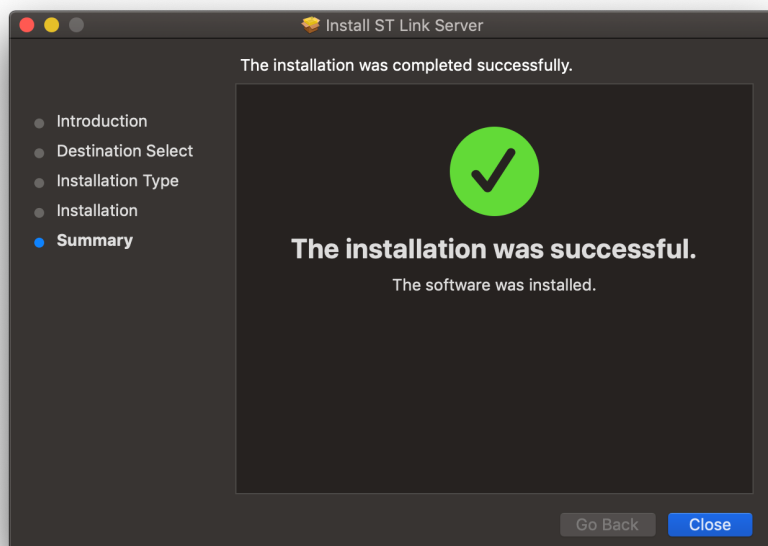
8. If a warning is displayed stating "This package is incompatible with this version of macOS and may fail to install", click on **[Install Anyway]**.

Figure 10. ST Link Server installer location dialog (macOS®)



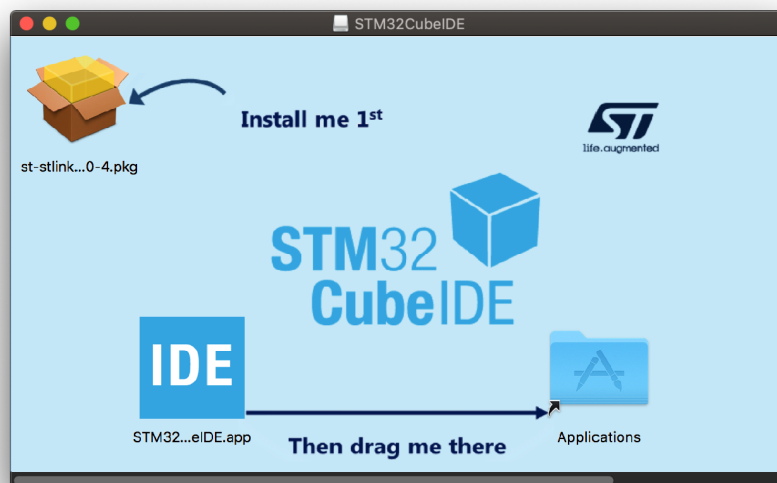
9. Click on **[Close]** and continue the installation of the STM32CubeIDE product.

Figure 11. ST Link Server installation finished (macOS®)



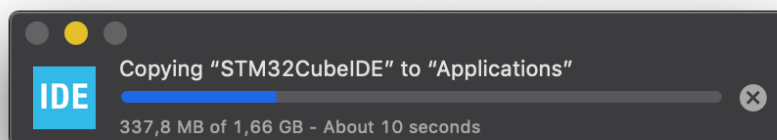
10. After successfully installing *ST Link Server*, drag the **STM32CubeIDE** icon to the **Applications** folder as indicated by the straight arrow.

Figure 12. STM32CubeIDE install page (macOS®)



11. Wait for the installation to finish. When done, it is possible to launch **STM32CubeIDE** from the **Launchpad** by clicking on the **IDE** icon.

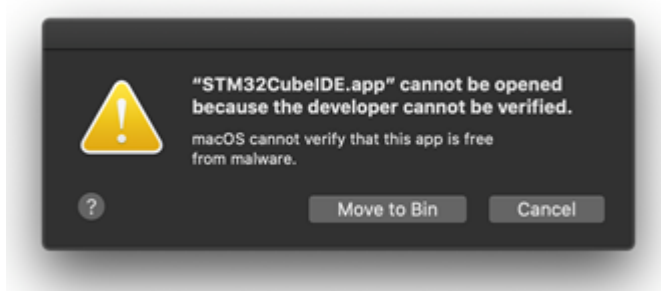
Figure 13. Installation progress bar (macOS®)



5.1 Gatekeeper tips

Because **STM32CubeIDE** is neither signed, nor Apple®-notarized, it is not unusual that the Gatekeeper blocks the first launch of the application as shown in [Figure 14](#).

Figure 14. macOS® - Gatekeeper blocking STM32CubeIDE launch



This section presents two workarounds for such a blocking issue, using either the [macOS security preferences](#) or the [Terminal application](#).

macOS® security preferences

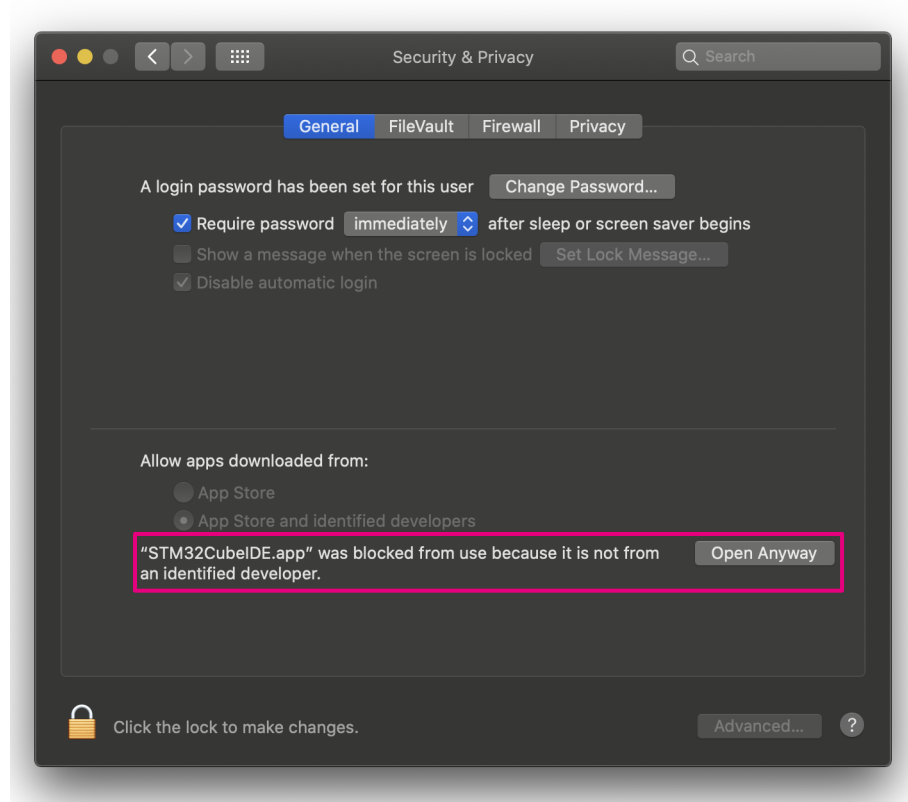
1. Open the *Security & Privacy* preferences in macOS® *System Preferences* (highlighted in [Figure 15](#)).

Figure 15. macOS® - System Preferences window



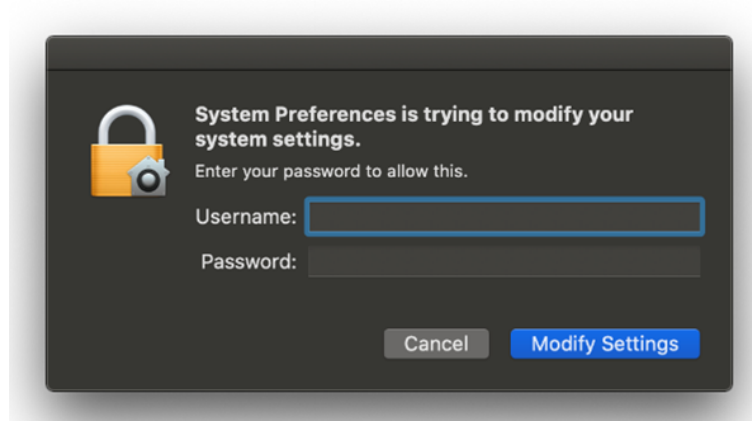
2. Click on **[Open Anyway]** (highlighted in Figure 16).

Figure 16. macOS® - Security & Privacy window



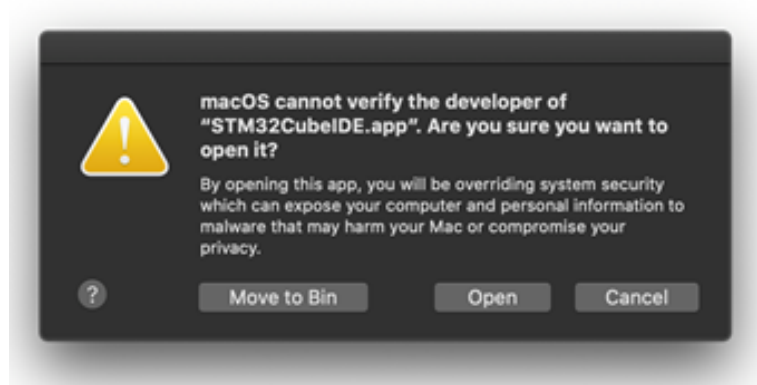
3. Enter administrator account credentials.

Figure 17. macOS® - Administrator authentication window



4. Confirm by clicking on **[Open]**.

Figure 18. macOS® - Opening a blocked application



Terminal application

Execute the following command:

```
xattr -c /Applications/STM32CubeIDE.app
```

5.2

Rosetta® installation on M1-based computers

To install **STM32CubeIDE** on an M1-based computer running macOS®, the user is prompted to install Rosetta® for the STM32CubeIDE installation to proceed.

After accepting to install Rosetta®, the user credentials must be entered (see [Figure 19](#)), after which Rosetta® installation starts (see [Figure 20](#)).

Figure 19. Authentication window for Rosetta® installation

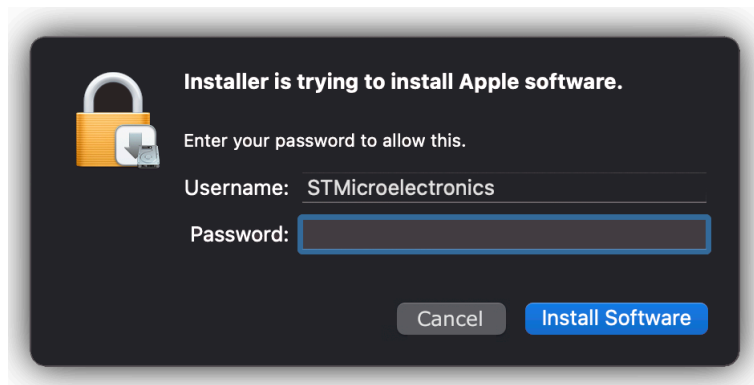
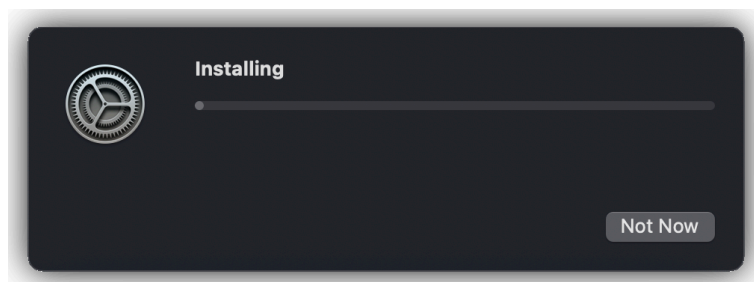


Figure 20. Rosetta® installation proceeding



6 Update an STM32CubeIDE installation

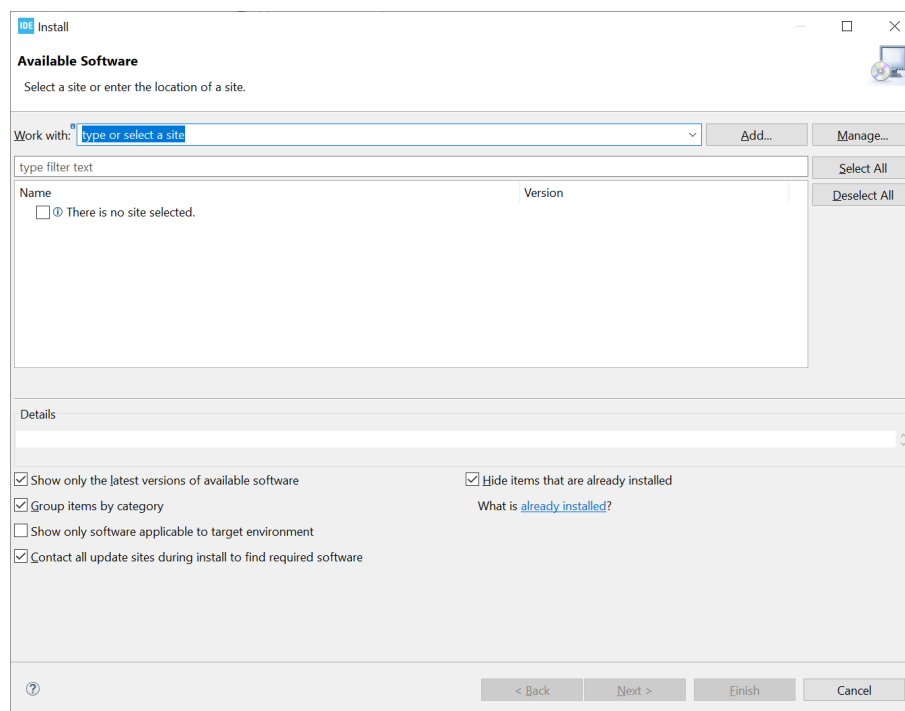
The Eclipse® update mechanism permits the quick update of available patches when STM32CubeIDE is already installed. To use this mechanism:

1. Launch STM32CubeIDE
2. Update the tool by **[Help]>[Check for updates...]**
3. Restart STM32CubeIDE
 - Either automatically if STM32CubeIDE proposes it
 - By exiting STM32CubeIDE and restarting it otherwise

By default, the update is done with the last version. If another version is needed, follow the steps below:

1. Open the *Install* dialog box as shown in [Figure 21](#) by **[Help]>[Install new software...]**

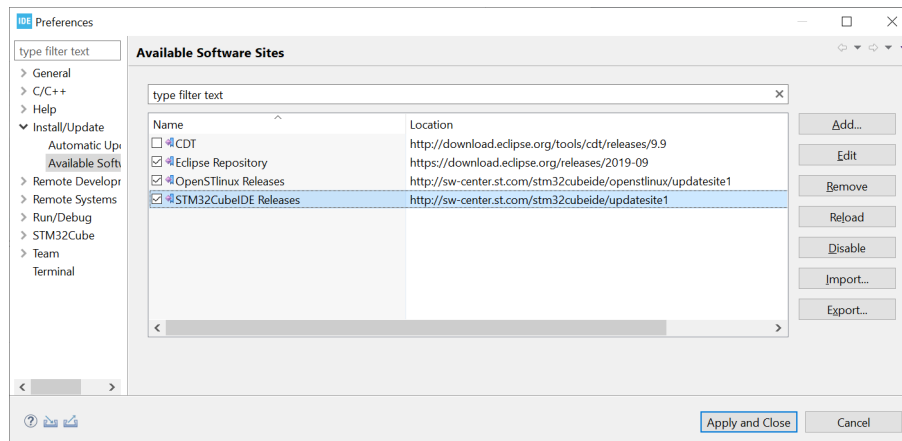
Figure 21. *Install* dialog box



2. Click on the **[Manage...]** button

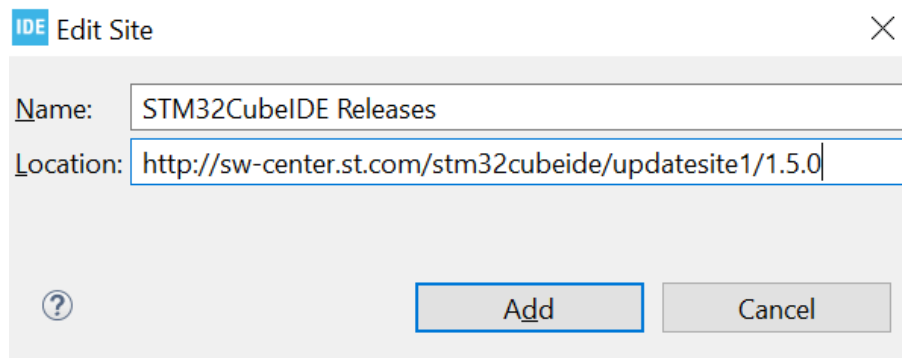
3. Select [STM32CubeIDE Releases] and click on [Edit] as shown in Figure 22

Figure 22. Preferences window



4. Add the desired version number at the end of the url in the [Location] field as shown in Figure 23

Figure 23. Edit Site dialog box



Note: The update mechanism has the same requirements as the installation in term of administration rights. If the installation has been made with administration rights, the update must be made also with the same administration rights.

7 Uninstall STM32CubeIDE (Windows®)

To uninstall STM32CubeIDE, under the installation folder:

1. Launch the product uninstaller (`uninstall.exe`)
2. Wait for the uninstaller dialog box and click on **[Uninstall]**

Note:

*In Windows®, the removal of the installation directory instead of using the uninstall procedure prevents further installation. In this case, delete registry key
`HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\STMicroelectronics\STM32CubeIDE\VersionToS`
uppress **using** `regedit.exe`.*

8 Uninstall STM32CubeIDE (Linux®)

The uninstallation of STM32CubeIDE depends on the distribution. Uninstall STM32CubeIDE according to the step below corresponding to the distribution used:

- For any distribution or if the distribution is not known:

```
sudo /opt/st/stm32cubeide_XXXX/uninstall.sh
```
- For RPM-based distributions (such as Red Hat®, CentOS™, SUSE®, Fedora® or others):

```
sudo rpm -e st-stm32cubeide_XXXX st-stlink-udev-rules st-stlink-server segger-jlink-udev-rules
```
- For Debian-based distributions (such as Debian®, Ubuntu® or others):

```
sudo apt-get remove st-stm32cubeide-XXXX st-stlink-udev-rules st-stlink-server segger-jlink-udev-rules
```

9 Uninstall STM32CubeIDE (macOS®)

To uninstall STM32CubeIDE:

1. Locate the version of STM32CubeIDE to uninstall in the *Applications* folder in the *Finder*
2. Drag the STM32CubeIDE app to uninstall to the trash
3. To delete the app permanently, choose **[Finder]>[Empty Trash]**

Revision history

Table 1. Document revision history

Date	Revision	Changes
18-Apr-2019	1	Initial version.
3-Nov-2020	2	Added: • <i>Update an STM32CubeIDE installation</i> • <i>Uninstall STM32CubeIDE (Windows)</i> • <i>Uninstall STM32CubeIDE (Linux)</i> • <i>Uninstall STM32CubeIDE (macOS)</i> Updated: • <i>System requirements</i> • <i>STM32CubeIDE installation (Linux)</i>
17-Nov-2021	3	Updated supported operating systems in sections <i>Microsoft Windows</i> and <i>macOS</i> . Added sections <i>Gatekeeper tips</i> and <i>Rosetta installation on M1-based computers</i> .
23-Feb-2022	4	Updated supported operating systems in sections Microsoft Windows and macOS .

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